EV EDISON · POWER EDISON — C3 SITE FINDER

Data Sources and Integration Report

November 4, 2026 · Based on an audit of the application code (commit 4637a0c), its configuration, and the candidate workbook `C3_candidate_sites_all_segments.xlsx` (built October 12, 2026; "Method and sources" sheet). C3 program facts and source links checked November 4, 2026.

16	2	30,584	0
sources built into the candidate workbook (incl. the NOFA and Power Edison sites)	map services the dashboard calls live (Google Maps, OpenStreetMap fallback)	candidate parcels imported and re-scored from the workbook	paid data vendors or enrichment APIs

How sources reach the dashboard. The government and open datasets below were combined into one Excel workbook before the application was built. The application imports that workbook into its own SQL Server database and recomputes every C3 score from it. It does **not** query those datasets live. While the dashboard runs, the only external services it calls are the Google Maps JavaScript API for the basemap, with OpenStreetMap tiles as a fallback. Street View and Google Maps are opened as ordinary links. Research the team records (screening decisions, feasibility checks, contacts) is entered in the app and is kept separate from the score.

1. How the data flows

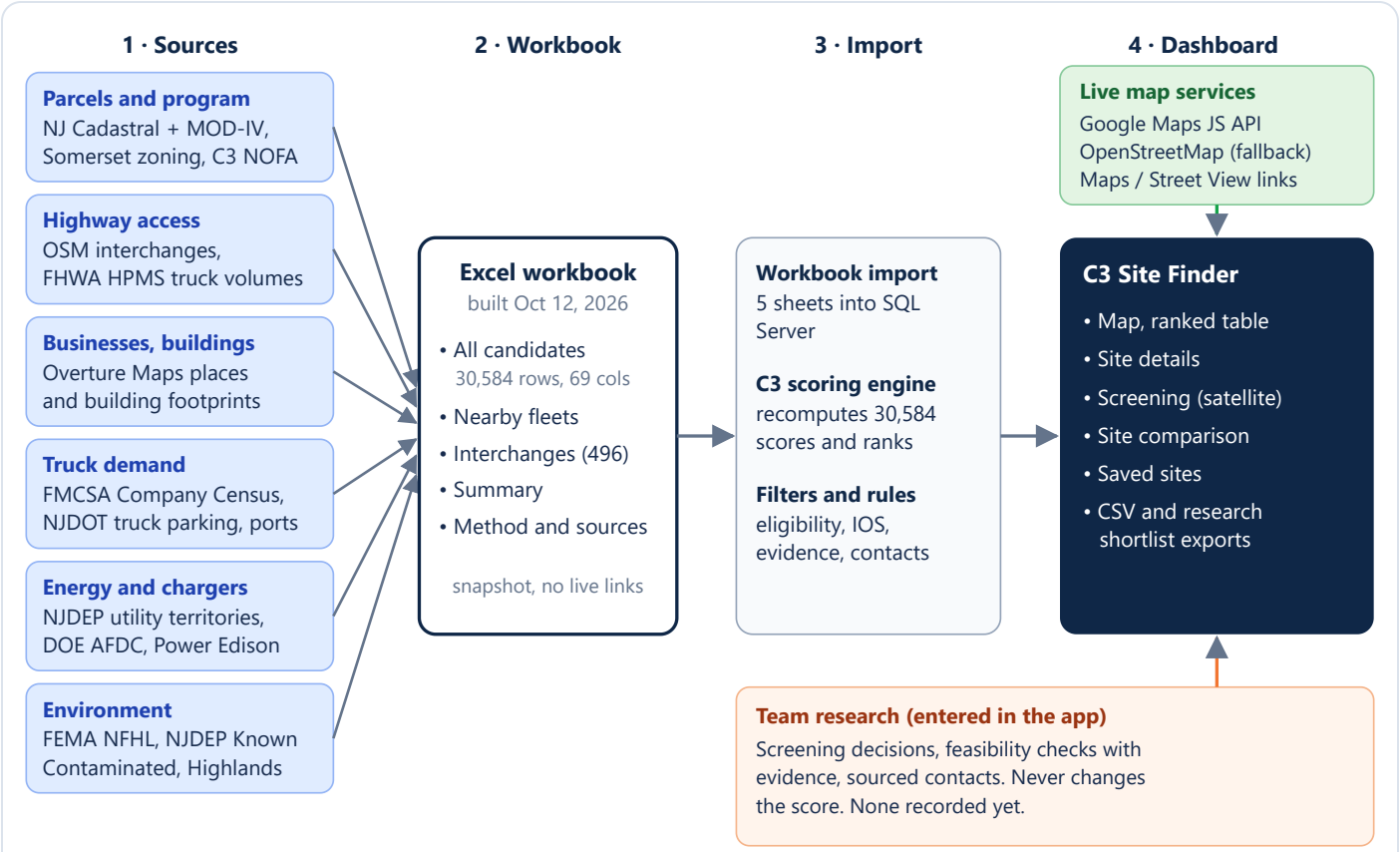


Figure 1. Upstream datasets were merged into the workbook outside this application. The app imports the workbook and re-scores it; while running, it calls only map services. Team research is stored separately and never changes the score.

2. C3 program requirements (checked against published sources)

Item	Published requirement	How the dashboard reflects it
Proposal deadline	December 23, 2026, 5:00 PM through NJDEP's C3 Infrastructure Program portal. Questions to C3Project@dep.nj.gov by November 30, 2026, 5:00 PM ET.	Countdown to December 23, 2026 on the dashboard.
Location	Within 5 miles of a designated interchange or exit on I-95 / NJ Turnpike, I-295 south of Exit 47, I-287, I-80, I-280 or I-78. The application portal asks for **"Distance to corridor in driving miles (Max. Five Miles)"**.	**Preliminary only.** The workbook and dashboard use *straight-line* miles to screen candidates (all within 5). Eligibility requires driving distance of 5 miles or less, which must be verified for each site; a site 5 straight-line miles away can be farther by road. **Exit 47 needs confirmation:** the workbook treats I-295 Exit 47 (47A/47B) as eligible, but the program describes eligible I-295 locations as south of Exit 47. 345 candidates (26 in a segment's top 100) use Exit 47A/47B as their nearest exit.
Charging power	At least 2.5 MW of charging per site and at least 250 kW per port.	**Not evaluated.** No capacity data is available, so the dashboard shows electrical capacity as Unknown until the utility confirms it.
Funding	Up to about $84 million for at least six charging depots, up to $20 million per site; funded by EPA's Climate Pollution Reduction Grants.	Context only; not used in scoring.
Segments	Six regional segments (NOFA Table 2), as recorded in the workbook.	Segment filter and per-segment ranking. *Not independently re-checked: the NOFA document itself could not be retrieved (NJDEP's site blocks automated access). Confirm against the NOFA before submission.*

Sources: NJDEP C3 Medium & Heavy-Duty EV Charging Infrastructure Program page · NJDEP Grant and Loan Programs · NJ League of Municipalities notice · EPA Climate Pollution Reduction Grants.

3. Sources built into the candidate workbook

Each source is confirmed by the workbook's *Method and sources* sheet and by the matching workbook columns the application imports. Links point to the provider's official page or service; where the workbook names no specific dataset, that is stated. **Workbook** means the source was used when the workbook was built; the dashboard uses the resulting columns.

	Source (as documented)	What it provides	How the dashboard uses it
1	**NJDEP C3 Medium & Heavy-Duty EV Charging NOFA** **Workbook** + app reference dep.nj.gov · C3 program page	Six application segments (NOFA Table 2), eligible interstates, Dec 23, 2026 deadline	Segment filter and per-segment ranking; eligibility screen; methodology page; proposal countdown
2	**NJ Cadastral parcels joined to MOD-IV tax data** (maps.nj.gov Framework/Cadastral) **Workbook** maps.nj.gov/.../Framework/Cadastral	Address, block/lot/qual, PAMS PIN, mapped and recorded acres, property class, assessed values, tax, last sale, owner mailing address, parcel shape, coordinates	Defines the 30,584 candidates; property class (8–15 pts), lot shape (−10 to +5); parcel card and mismatch signals in Site Details; owner mailing address

	Source (as documented)	What it provides	How the dashboard uses it
3	**Somerset County parcel layer** Workbook No URL recorded in the workbook	Zoning codes (Somerset County only)	Zoning shown in Site Details where available
4	**OpenStreetMap** motorway_junction nodes in NJ Workbook OSM: highway=motorway_junction openstreetmap.org/copyright (ODbL)	496 eligible interchanges; nearest exit and straight-line miles	Preliminary 5-mile straight-line screen (driving distance still to verify); exit distance (20 pts); interstate filter; "within 1 mi" preference
5	**Overture Maps places** (release 2026-09-23.1) Workbook docs.overturemaps.org	Up to 3 businesses inside each parcel, categories, listed phone and website; freight / industrial flags; truck stops	Trucking on site (10 / 4 pts); busy multi-tenant (−5); freight filter; IOS classification; business card (listed, not confirmed on site)
6	**Overture Maps building footprints** Workbook docs.overturemaps.org	Open acres (parcel minus buildings), building coverage, warehouses ≥ 200,000 sq ft within 2 mi	Open land (−15 to +20); large warehouses (6); preliminary parking estimate (~15 trailers per open acre)
7	**FMCSA Company Census** (data.transportation.gov az4n-8mr2) Workbook data.transportation.gov/d/az4n-8mr2	Active carriers in NJ and nearby PA/NY/DE ZIPs: trucks, tractors and carriers within 3/5/10 mi; 10 largest fleets near each top-100 site (6,000 rows)	Fleets based nearby (12 pts); "Nearby carriers" table, labelled as *not confirmed on site*
8	**FHWA HPMS 2023, New Jersey** (geo.dot.gov) Workbook geo.dot.gov · HPMS_FULL_NJ_2023 fhwa.dot.gov · HPMS	Tractor-trailer AADT next to the site and at the exit; distance to roads carrying 500+ tractor-trailers/day	Truck traffic (10 pts); far-from-truck-road penalty (−10); Trucks/day column and sort
9	**NJDOT large-truck parking layer** (56 sites) Workbook No URL recorded in the workbook	Truck stops / parking within 2 mi; distance to the nearest	Truck stop or parking (3 pts)
10	**Port terminal locations** (6 approximate points) Workbook Compiled points; no external dataset	Straight-line miles to the nearest port	Port within 5 / 10 mi (4 / 2 pts)
11	**NJDEP Electric Utilities Territory Map** Workbook mapsdep.nj.gov · Utilities layer 10	Electric utility serving each parcel	Utility filter; power context in At a glance (territory only, says nothing about capacity)
12	**DOE AFDC station locator** (DC fast, NJ, open + planned) Workbook afdc.energy.gov/stations	Nearest DC fast charger and medium/heavy-duty-capable charger distances	No truck charger within 10 mi (up to 10 pts); no DCFC within 2 mi (5)

	Source (as documented)	What it provides	How the dashboard uses it
13	**FEMA NFHL** 100-year flood zones Workbook fema.gov · National Flood Hazard Layer	Flood zone (checked for the top 150 sites per segment)	Flood penalty (−10); flag; exclude and "flood-checked only" filters
14	**NJDEP Known Contaminated Sites** Workbook dep.nj.gov/srp/kcsnj mapsdep.nj.gov · KCS layer	Known contaminated sites (top 150 per segment)	Flag and exclude filter (not part of the score)
15	**Highlands Preservation / Planning Areas** Workbook nj.gov/njhighlands (Highlands Council)	Highlands designation (the workbook does not name the publisher; the Highlands Council is the regional planning body)	Preservation Area penalty (−15); exclude filter
16	**Power Edison sites** (company list: Kearny Point, Irvington, Edison, Paterson) Workbook + app seed Internal company data	Nearest Power Edison site and distance	Map layer; "within 2 mi" flag and exclude filter (NJDEP may fund only one of two nearby proposals)

4. Services the dashboard uses while running

Service	Type	How it is integrated and used
Google Maps JavaScript API (maps and marker libraries, weekly channel) developers.google.com/maps	Live API	Loaded with `@googlemaps/js-api-loader`. The key is supplied from .NET user-secrets through `/api/config/maps` and has been removed from the current tracked configuration. **Open items:** the key was committed in earlier local commits, so rotating it in Google Cloud and cleaning the local Git history are still pending. Provides the dashboard basemap (road and satellite views) under the site points drawn with deck.gl, and the satellite view with a site marker on the Screening page. Display only: no Google data is stored.
OpenStreetMap standard tiles (tile.openstreetmap.org) OSM tile usage policy	Live tiles	Fallback basemap used only when no Google key is configured, with "© OpenStreetMap contributors" attribution. Suitable for local or demo use, not heavy production traffic.
Google Maps and Street View links	Link-out	"Satellite", "Google Maps" and "Street View" buttons in Site Details and Screening open the site's coordinates in a new tab. No API call and no key.
Google Fonts (Inter)	UI only	Typeface for the interface. Not a data source.

5. Sources investigated but not integrated

These sources appear nowhere in the application code, its configuration, or the workbook's columns and methodology.

Source	Status	Finding
EPA ECHO	Not integrated	No reference. Contamination comes from NJDEP Known Contaminated Sites instead.
NJTPA data	Not integrated	No reference. Truck volumes come from FHWA HPMS; NJDOT's large-truck network is not published as GIS, so HPMS roads with 500+ tractor-trailers/day are the stand-in.
FMCSA SAFER (carrier lookup)	Not integrated	No live SAFER lookups. Carrier data comes from the bulk FMCSA Company Census as of the workbook date.

Source	Status	Finding
NJGIN aerial imagery	Not integrated	NJ parcel boundaries are used (maps.nj.gov Cadastral), but NJ orthoimagery is not. Imagery review uses Google satellite views.
Google Places API	Not integrated	No Places calls. Business names, categories, phones and websites come from Overture Maps places.
Google Street View API	Not integrated	Street View is opened only as a link (Section 4); no imagery is embedded or downloaded.
DOE AFDC API (live)	Indirect only	AFDC station data is used through the workbook snapshot (Section 3, item 12); the app makes no AFDC calls.
Paid data vendors / enrichment APIs	None	No paid enrichment was used. Owner names remain blank (see limitations).

6. Limitations to keep in mind

- **Upstream build not in this repository.** The scripts that assembled the workbook are not part of the project. The 16 sources above are confirmed from the workbook's own documentation and columns, not by re-running that build.
- **C3 power minimums not checked.** The program's 2.5 MW per site and 250 kW per port minimums cannot be assessed from the data; each shortlisted site needs a utility capacity check.
- **Snapshots, not live feeds.** Every workbook source reflects its extract date (e.g. Overture release 2026-09-23.1, HPMS 2023). Refreshing means importing a new workbook; the app then re-scores and keeps the team's research linked by Site ID.
- **Partial checks.** Flood zones and contaminated sites were checked only for the top 150 sites per segment; other sites show "not checked".
- **Straight-line screening vs driving-distance eligibility.** All exit distances in the workbook and dashboard are straight-line and serve only as a preliminary screen. The C3 application requires the distance to the corridor in driving miles (maximum 5), so every shortlisted site needs a driving-distance check; sites far from an exit are most at risk (2,536 candidates are over 4 straight-line miles away). Sites using I-295 Exit 47A/47B also need eligibility confirmed.
- **Owner names.** Blank in every public NJ parcel service (2020 law A6171, related to Daniel's Law). Use the municipal tax assessor, an OPRA request to the county tax board, or a vendor.
- **Listings are not confirmations.** Businesses are matched to parcels by location, and carriers are placed at their registered ZIP centroid. Neither confirms who operates on a property.
- **Verification comes only from the team.** "Verified" in the dashboard means a reviewer recorded evidence (and, for contacts, a source). A high C3 score never verifies a site. As of this report, no research has been recorded, so all sites read "Unknown".

Prepared from the application code and configuration (commit 4637a0c) and the workbook's "Method and sources" sheet (built 2026-10-12). Program facts and links checked November 4, 2026; driving-mile requirement verified by the team on the NJDEP C3 application portal.